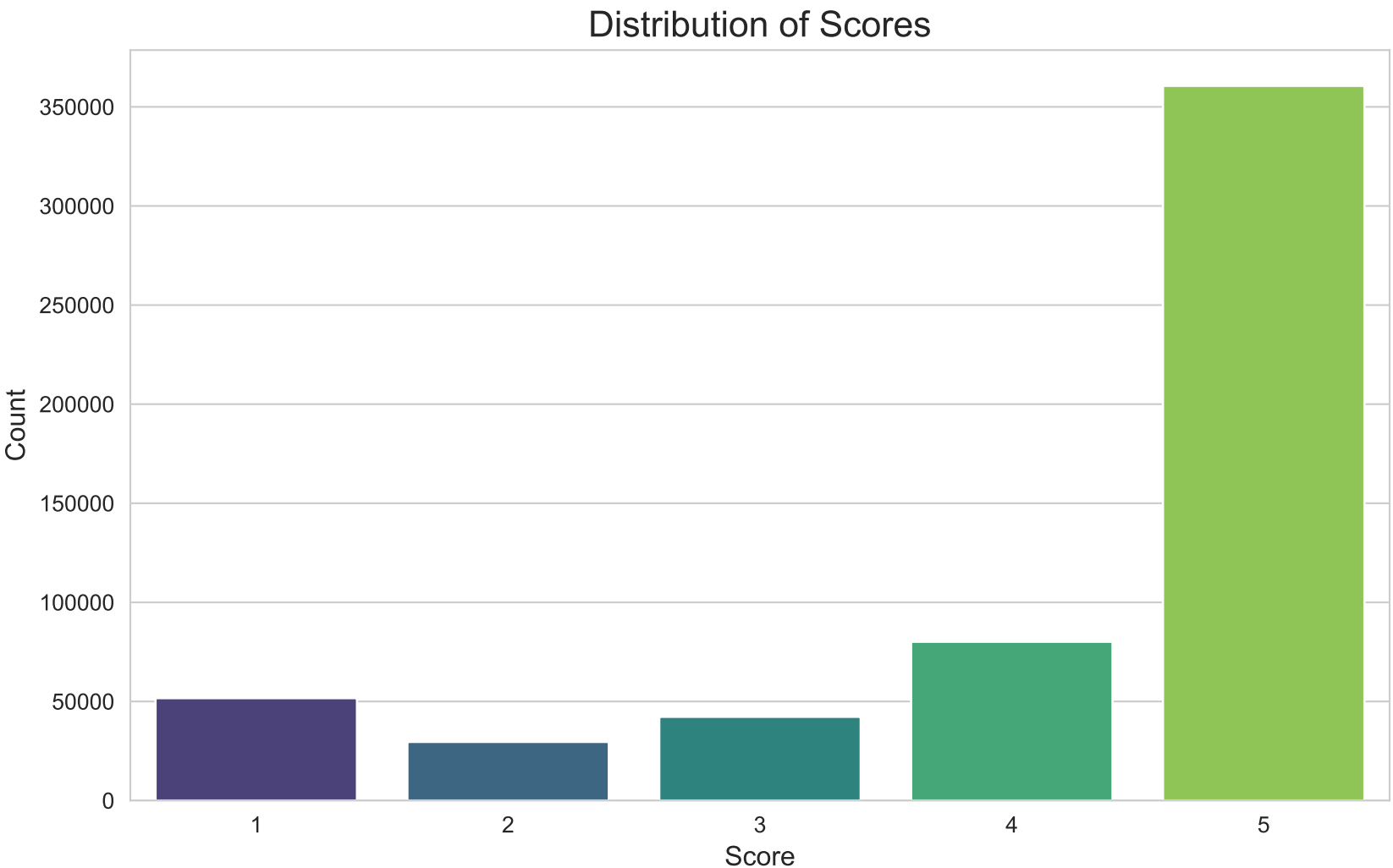
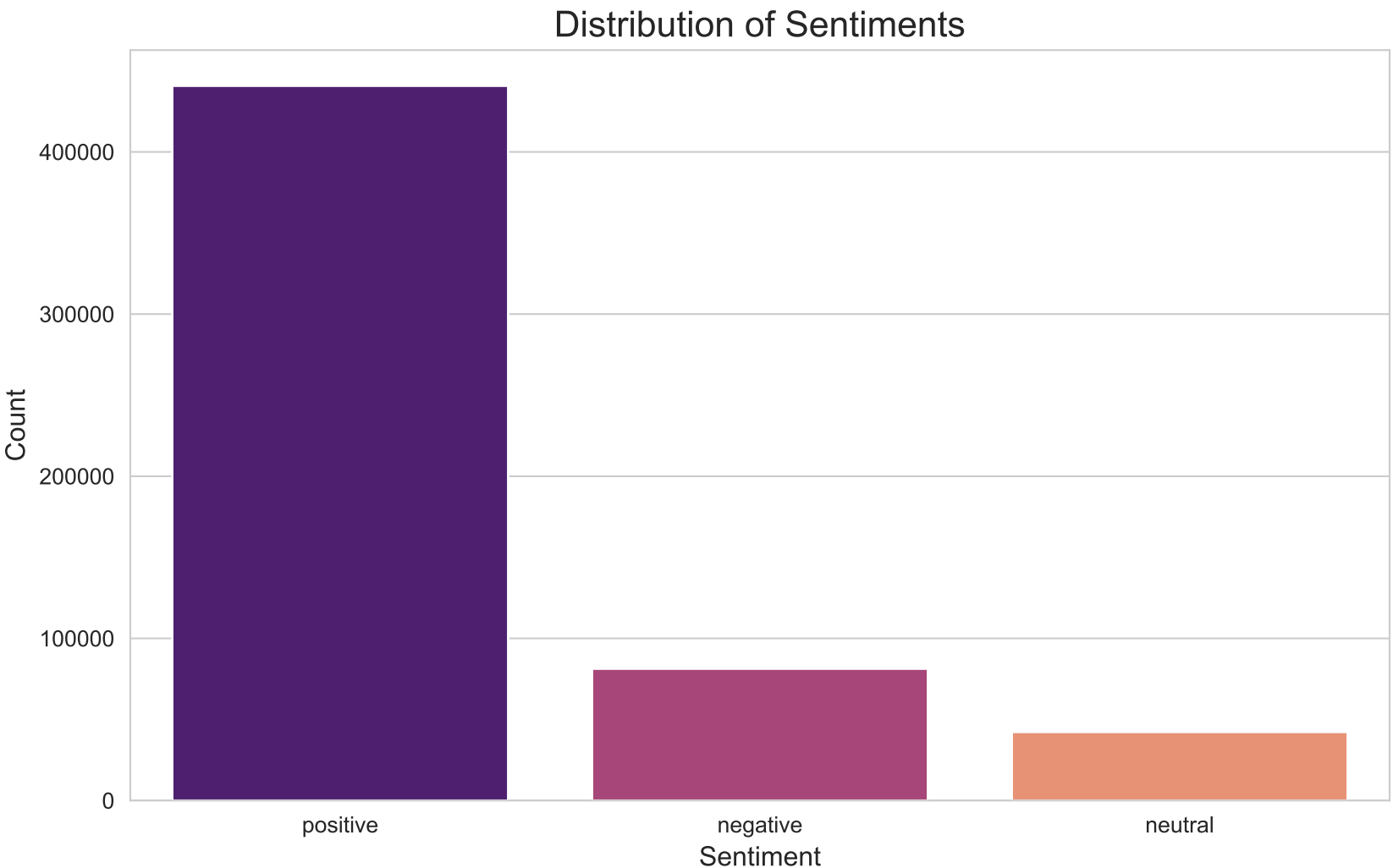
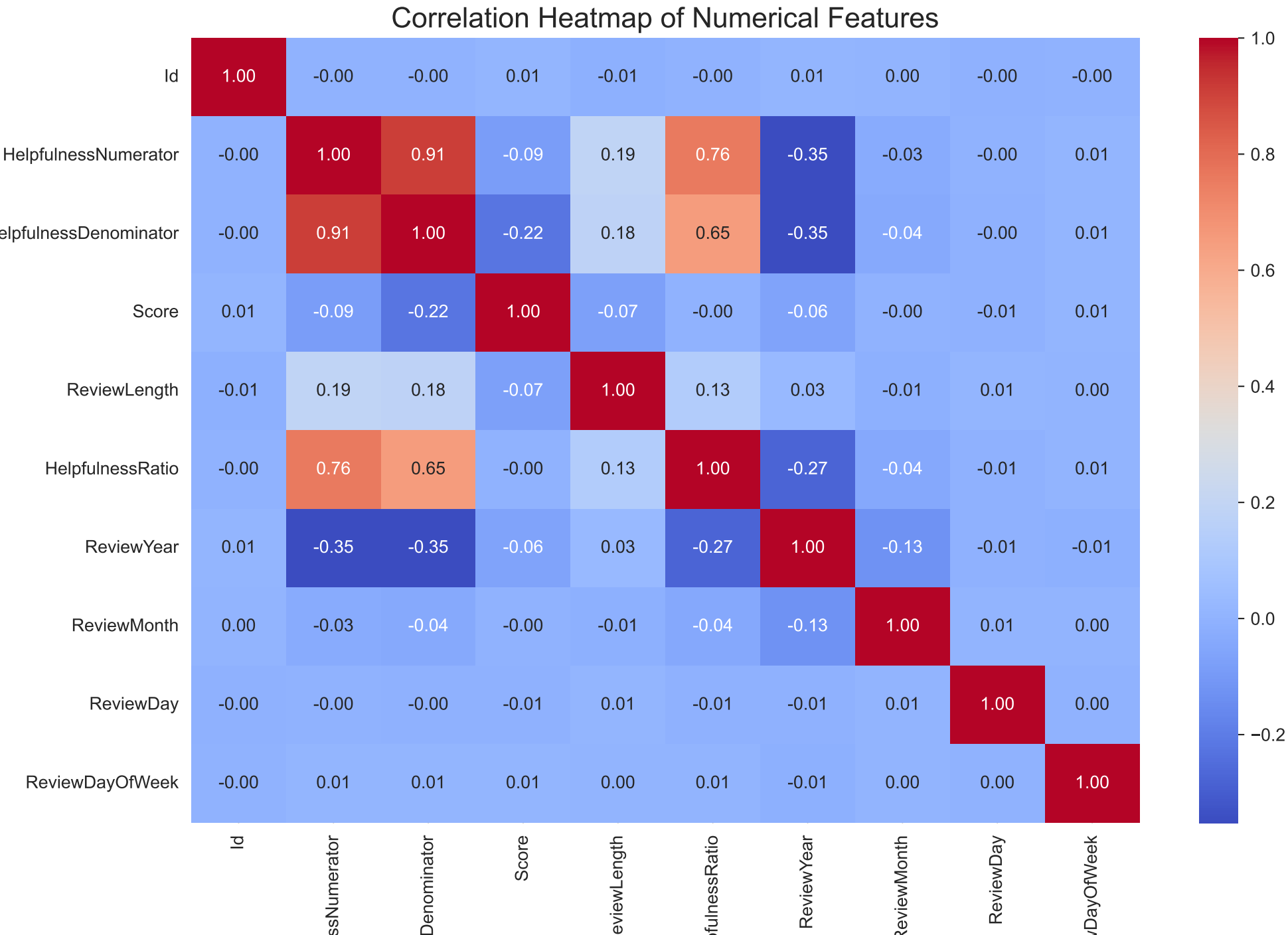
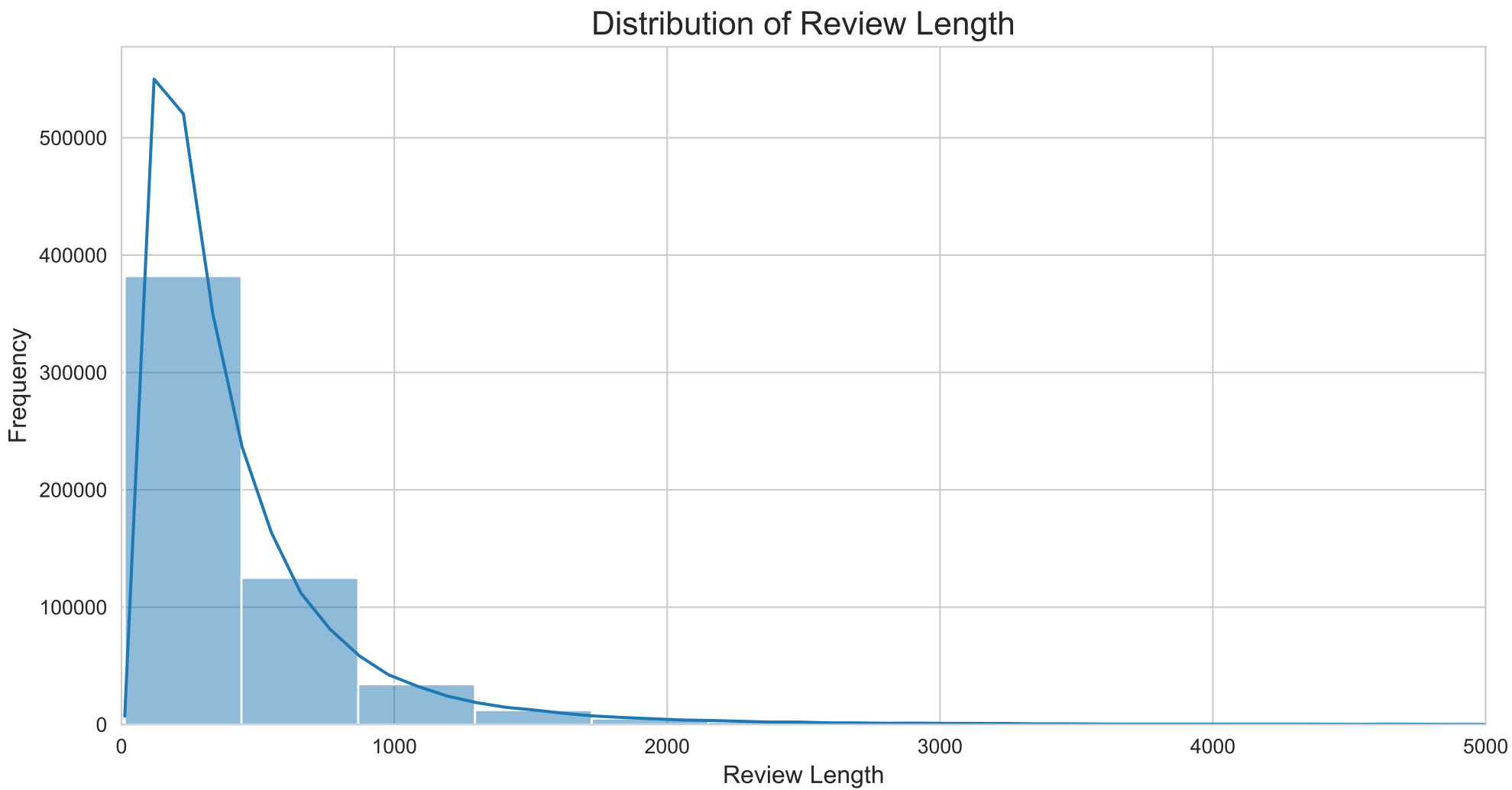
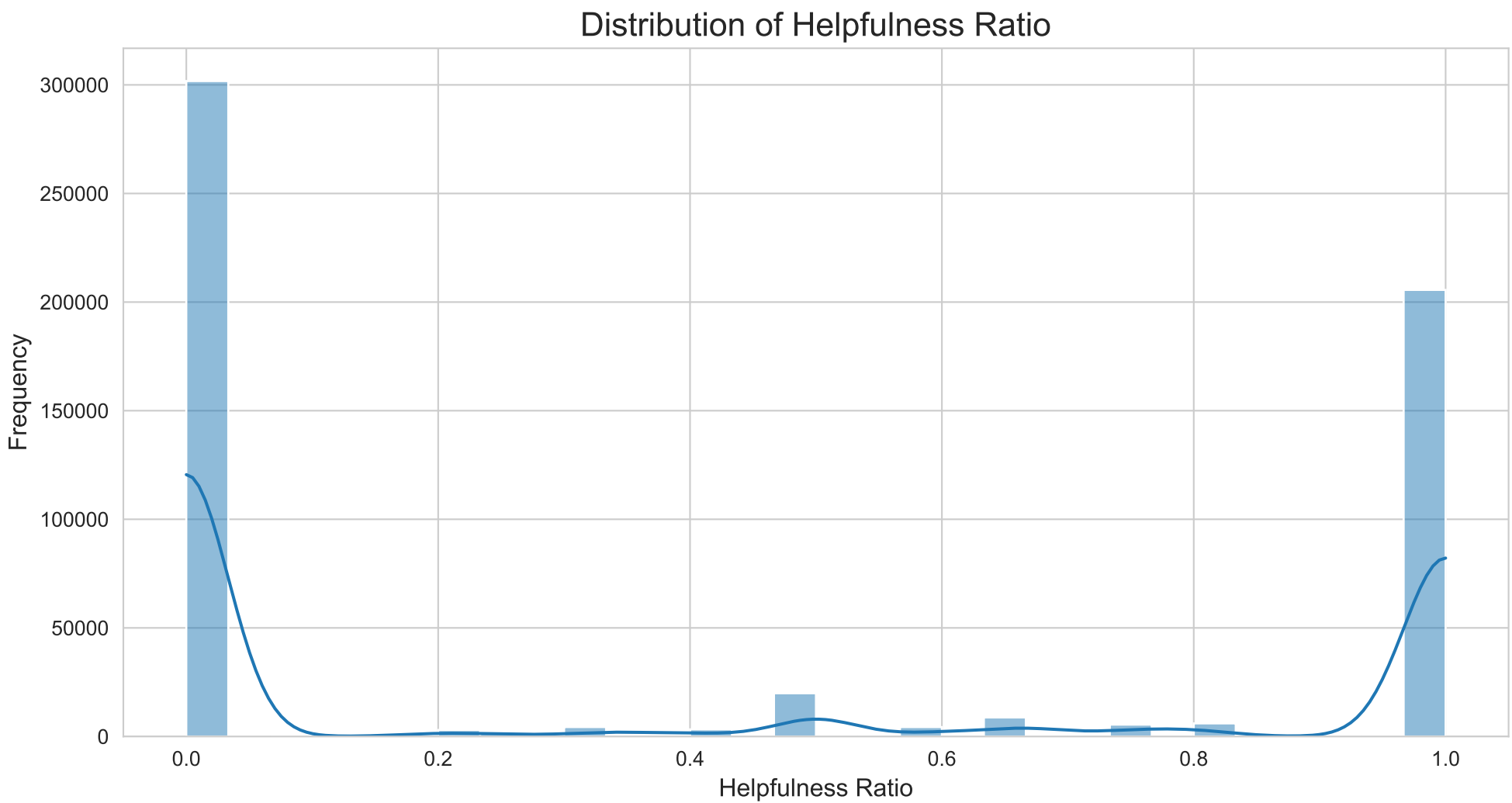
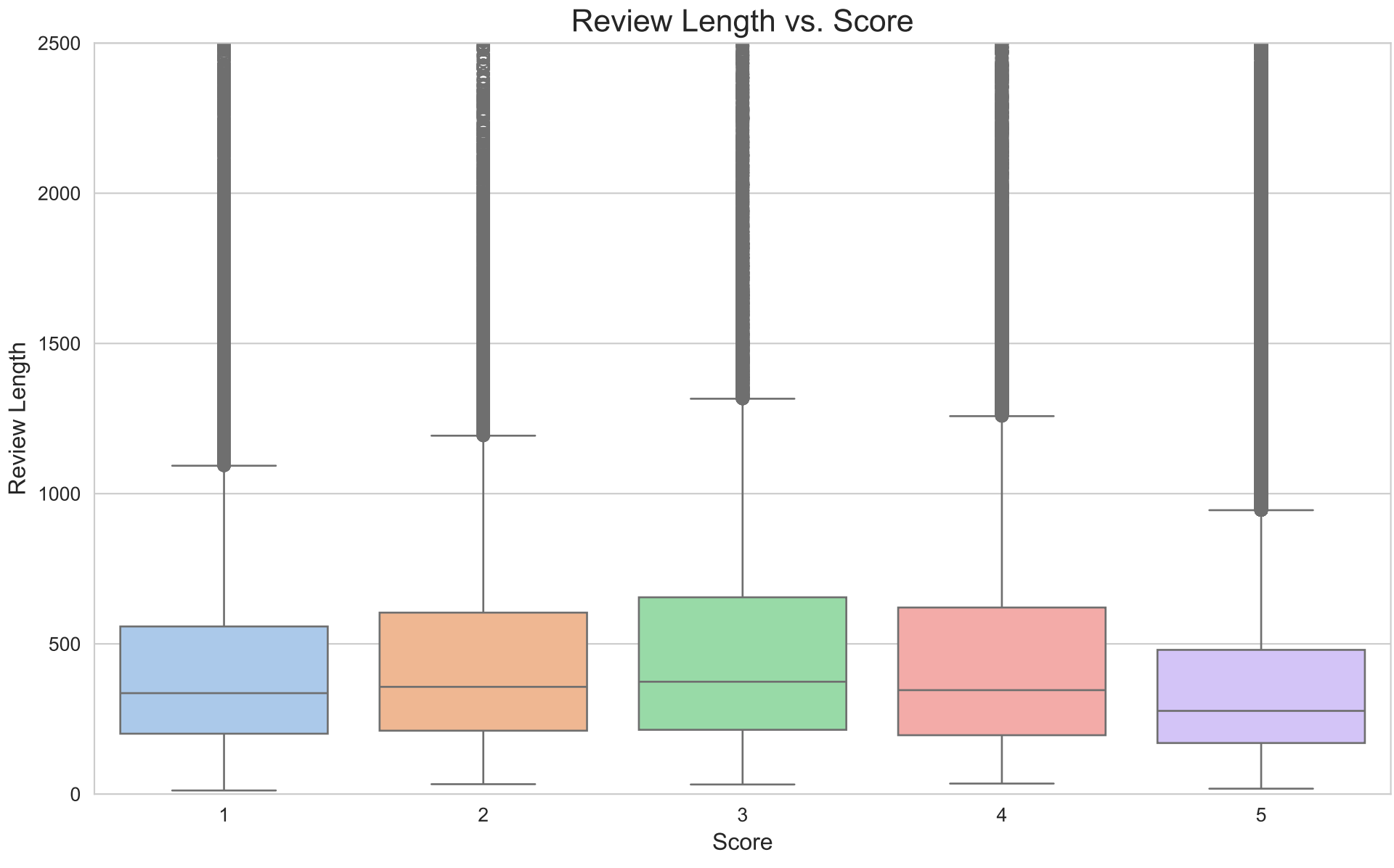

Distribution of Scores
Count
350000
300000
250000
200000
150000
100000
50000
0
1
2
3
4
5
Score

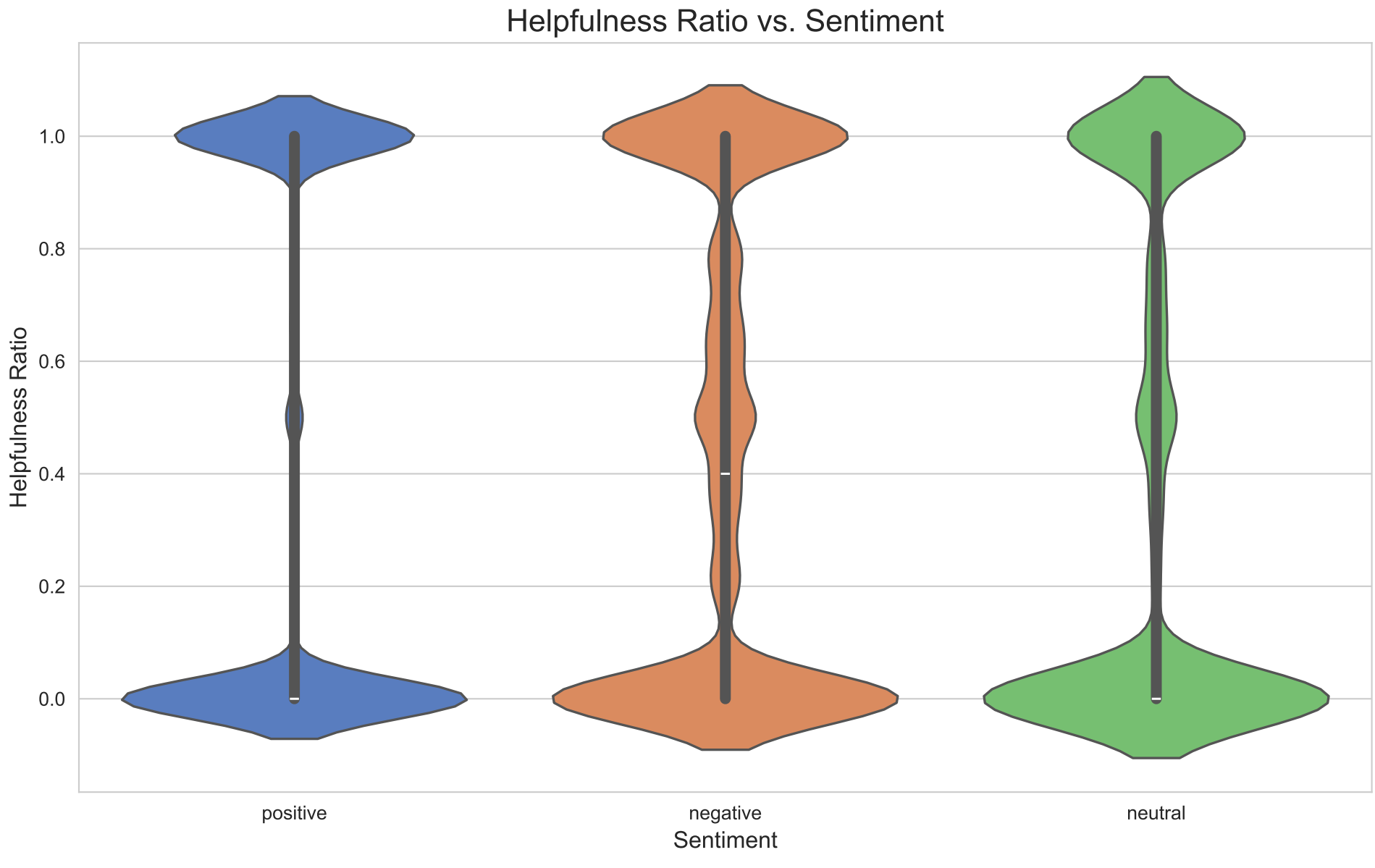

Distribution of Sentiments
400000
300000
200000
100000
0
Count
positive
negative
neutral
Sentiment

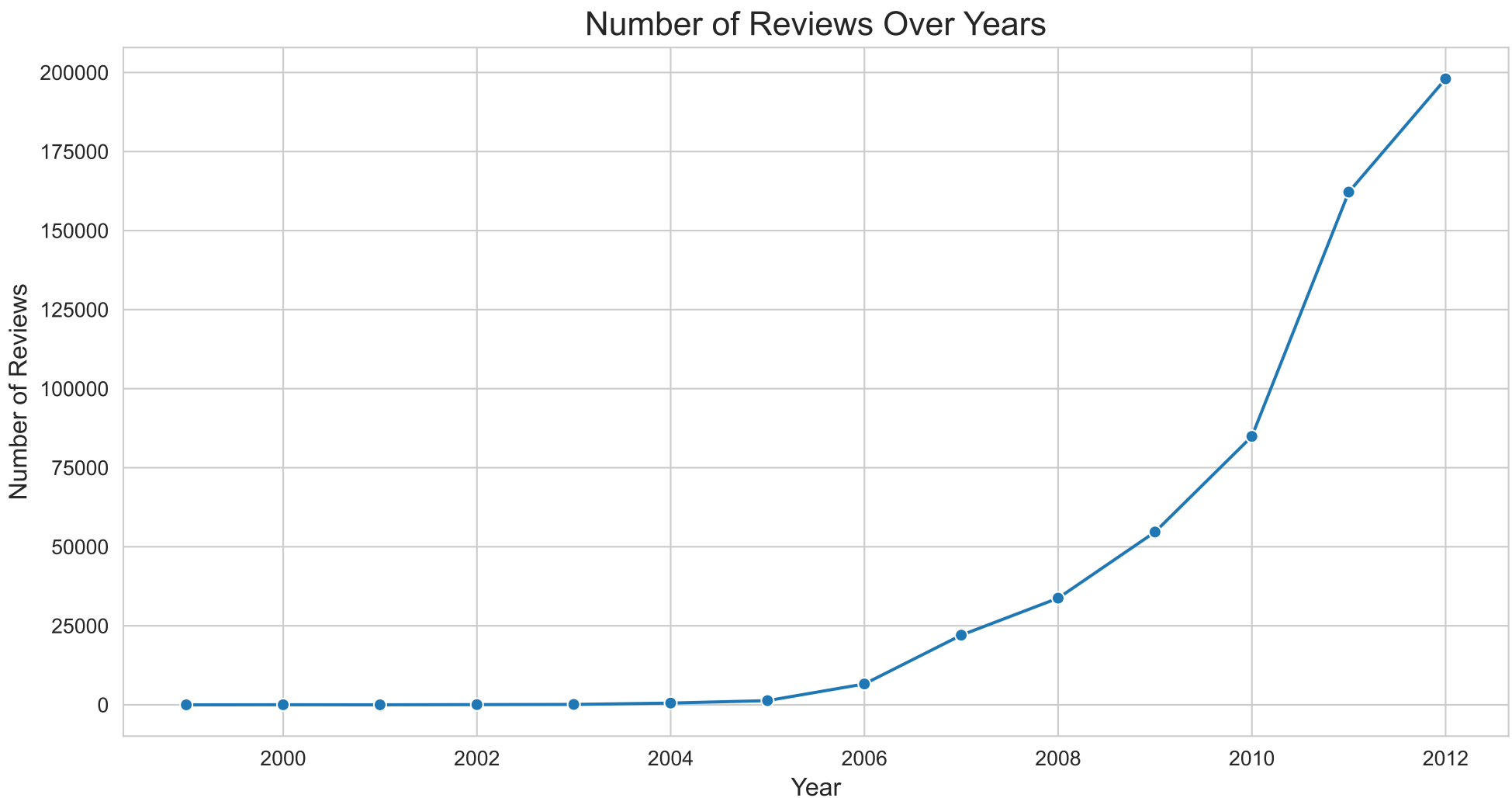

Correlation Heatmap of Numerical Features
Id
HelpfulnessNumerator
elpfulnessDenominator
Score
ReviewLength
HelpfulnessRatio
ReviewYear
ReviewMonth
ReviewDay
ReviewDayOfWeek
1.00 -0.00 -0.00 0.01 -0.01 -0.00 0.01 0.00 -0.00 -0.00
-0.00 1.00 0.91 -0.09 0.19 0.76 -0.35 -0.03 -0.00 0.01
-0.00 0.91 1.00 -0.22 0.18 0.65 -0.35 -0.04 -0.00 0.01
0.01 -0.09 -0.22 1.00 -0.07 -0.00 -0.06 -0.00 -0.01 0.01
-0.01 0.19 0.18 -0.07 1.00 0.13 0.03 -0.01 0.01 0.00
-0.00 0.76 0.65 -0.00 0.13 1.00 -0.27 -0.04 -0.01 0.01
0.01 -0.35 -0.35 -0.06 0.03 -0.27 1.00 -0.13 -0.01 -0.01
0.00 -0.03 -0.04 -0.00 -0.01 -0.04 -0.13 1.00 0.01 0.00
-0.00 -0.00 -0.00 -0.01 0.01 -0.01 -0.01 0.01 1.00 0.00
-0.00 0.01 0.01 0.01 0.00 0.01 -0.01 0.00 0.00 1.00
ssNumerator
Denominator
eviewLength
fulnessRatio
ReviewYear
ReviewMonth
ReviewDay
wDayOfWeek
1.0
0.8
0.6
0.4
0.2
0.0
−0.2

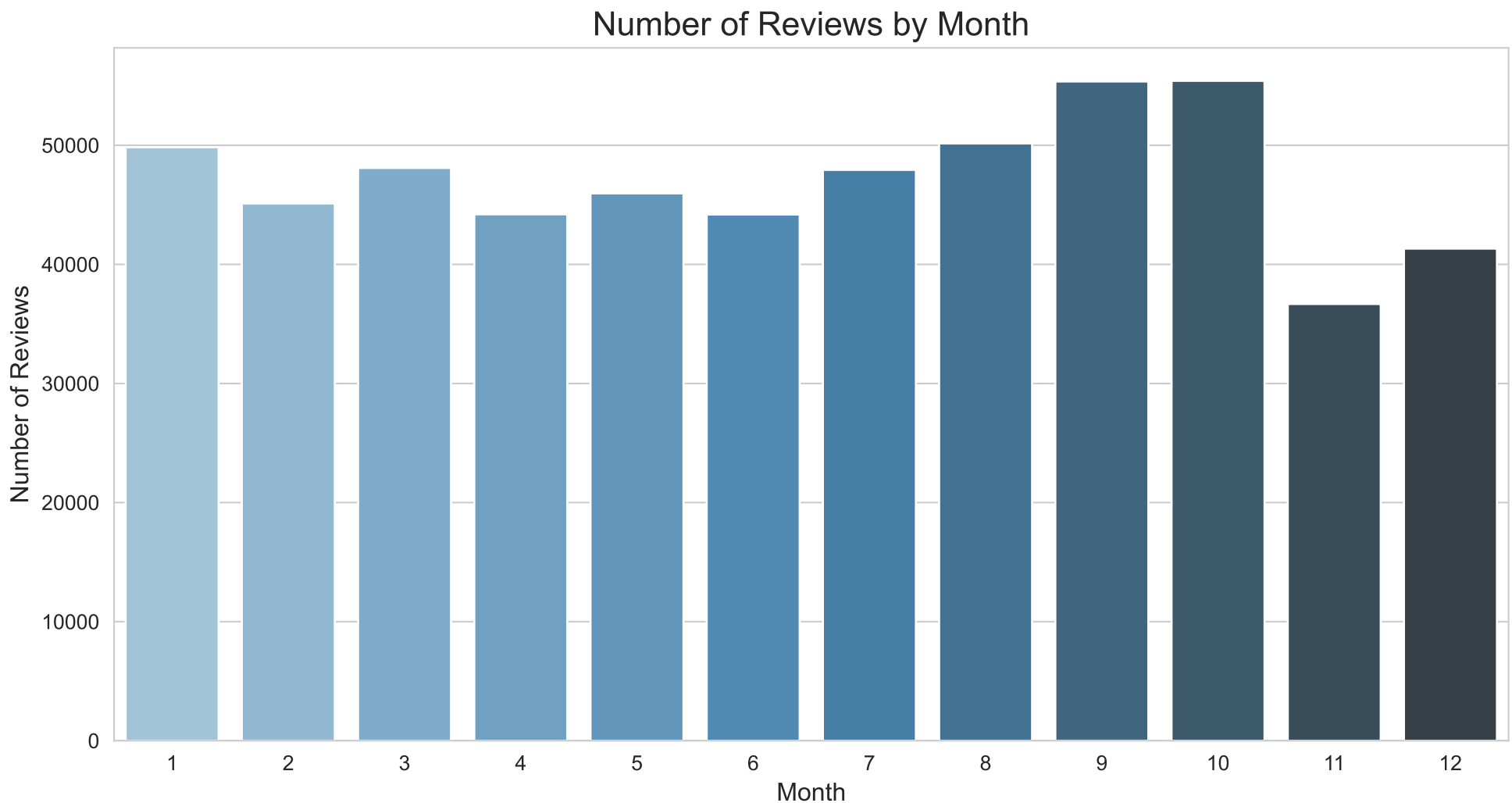

Distribution of Review Length
Frequency
500000
400000
300000
200000
100000
0
0
1000
2000
3000
4000
5000
Review Length

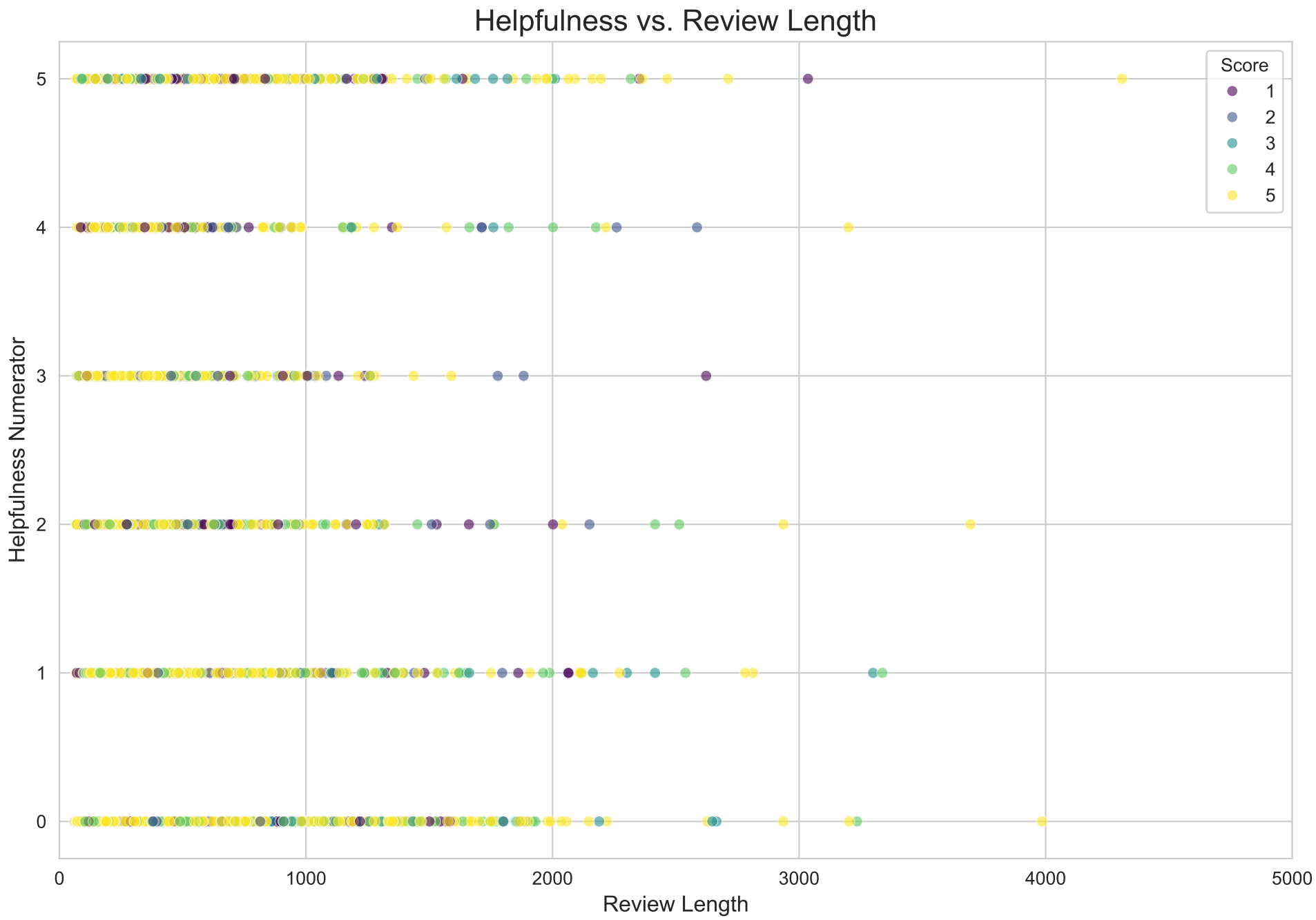

Distribution of Helpfulness Ratio
Frequency
300000
250000
200000
150000
100000
50000
0
0.0
0.2
0.4
0.6
0.8
1.0
Helpfulness Ratio

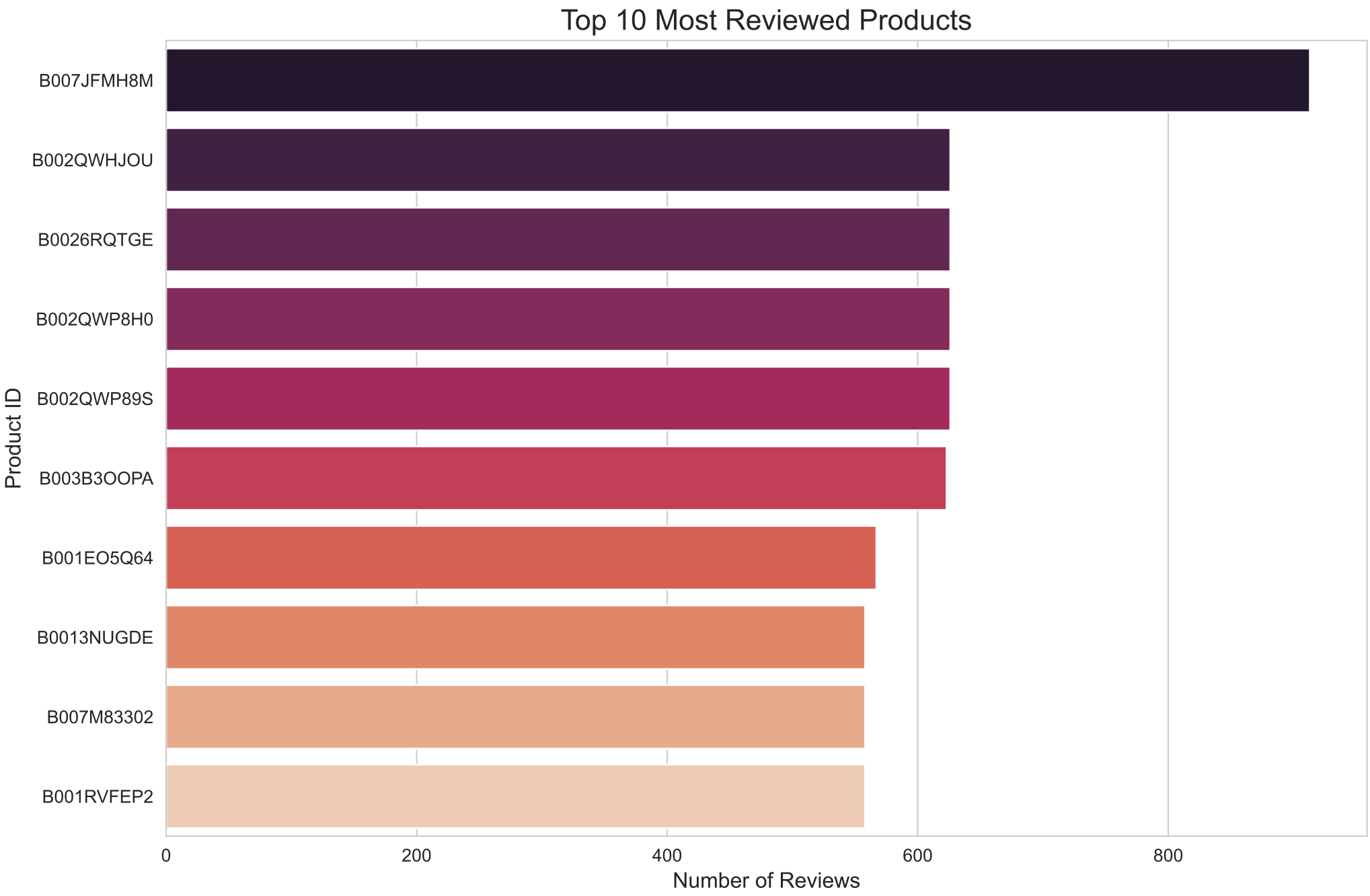

Review Length vs. Score
Review Length
2500
2000
1500
1000
500
0
1
2
3
4
5
Score

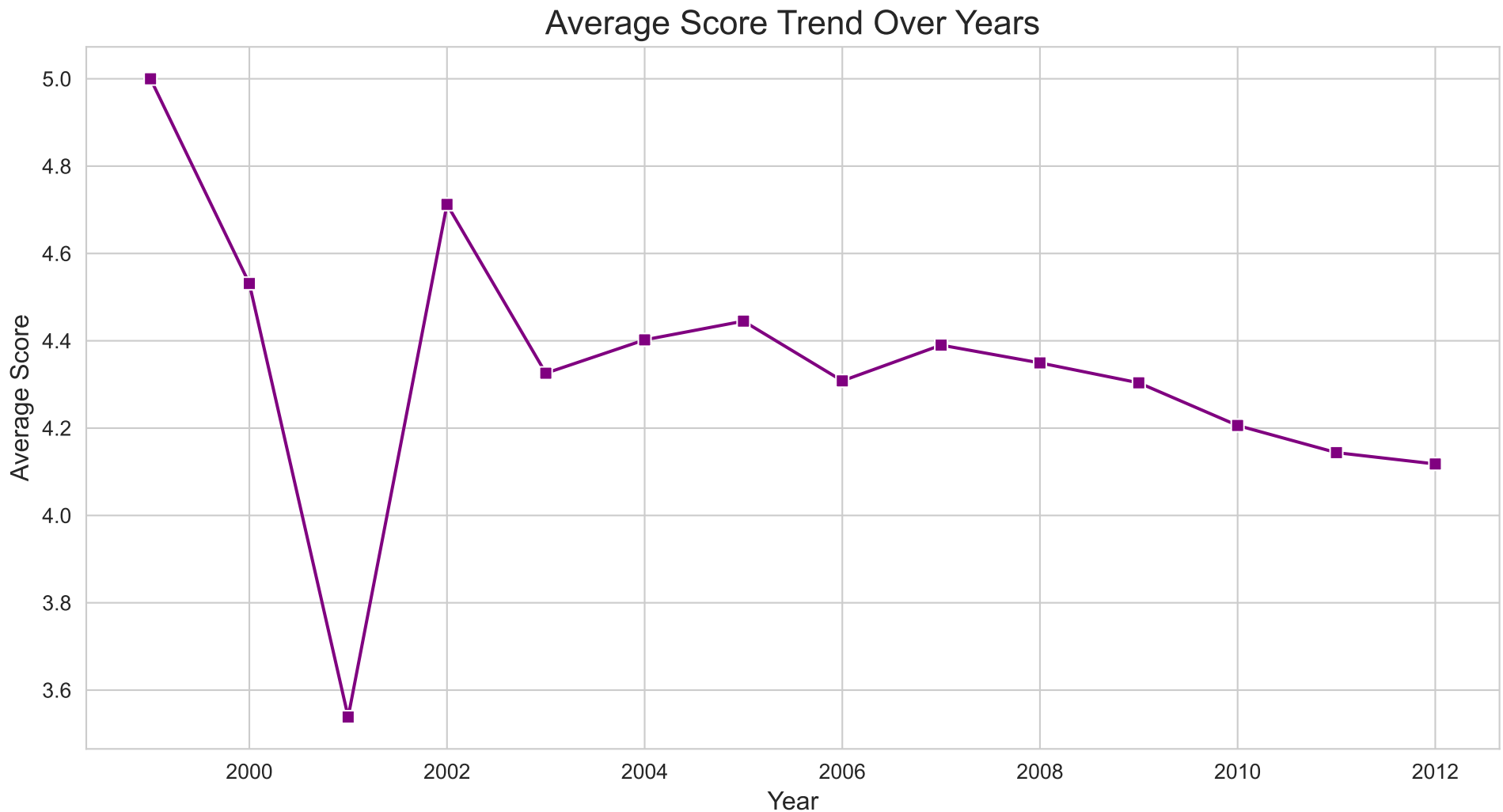

Helpfulness Ratio vs. Sentiment
Helpfulness Ratio
1.0
0.8
0.6
0.4
0.2
0.0
positive
negative
neutral
Sentiment


Number of Reviews Over Years
Number of Reviews
200000
175000
150000
125000
100000
75000
50000
25000
0
2000
2002
2004
2006
2008
2010
2012
Year

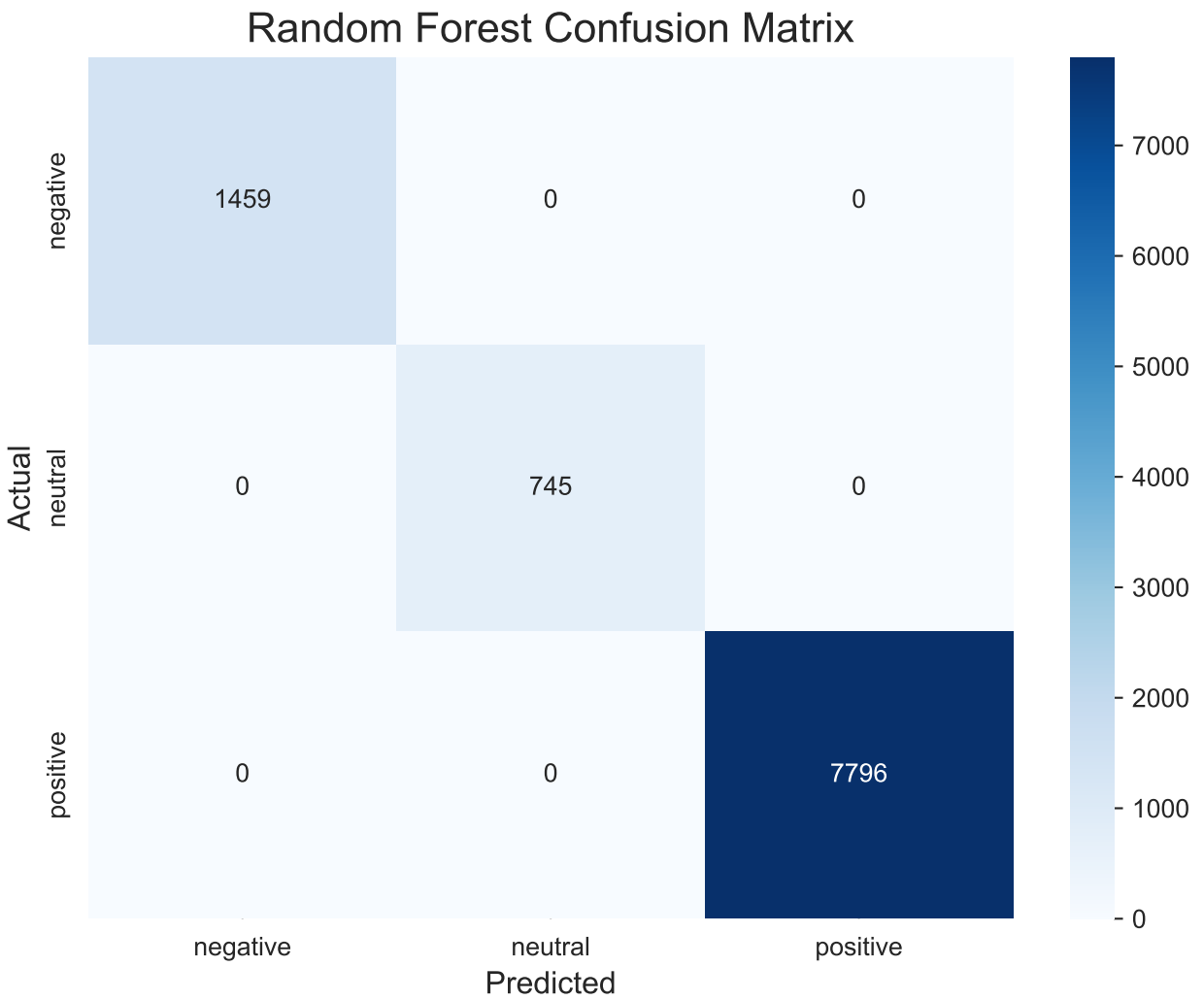

Number of Reviews by Month
Number of Reviews
0
10000
20000
30000
40000
50000
1
2
3
4
5
6
7
8
9
10
11
12
Month

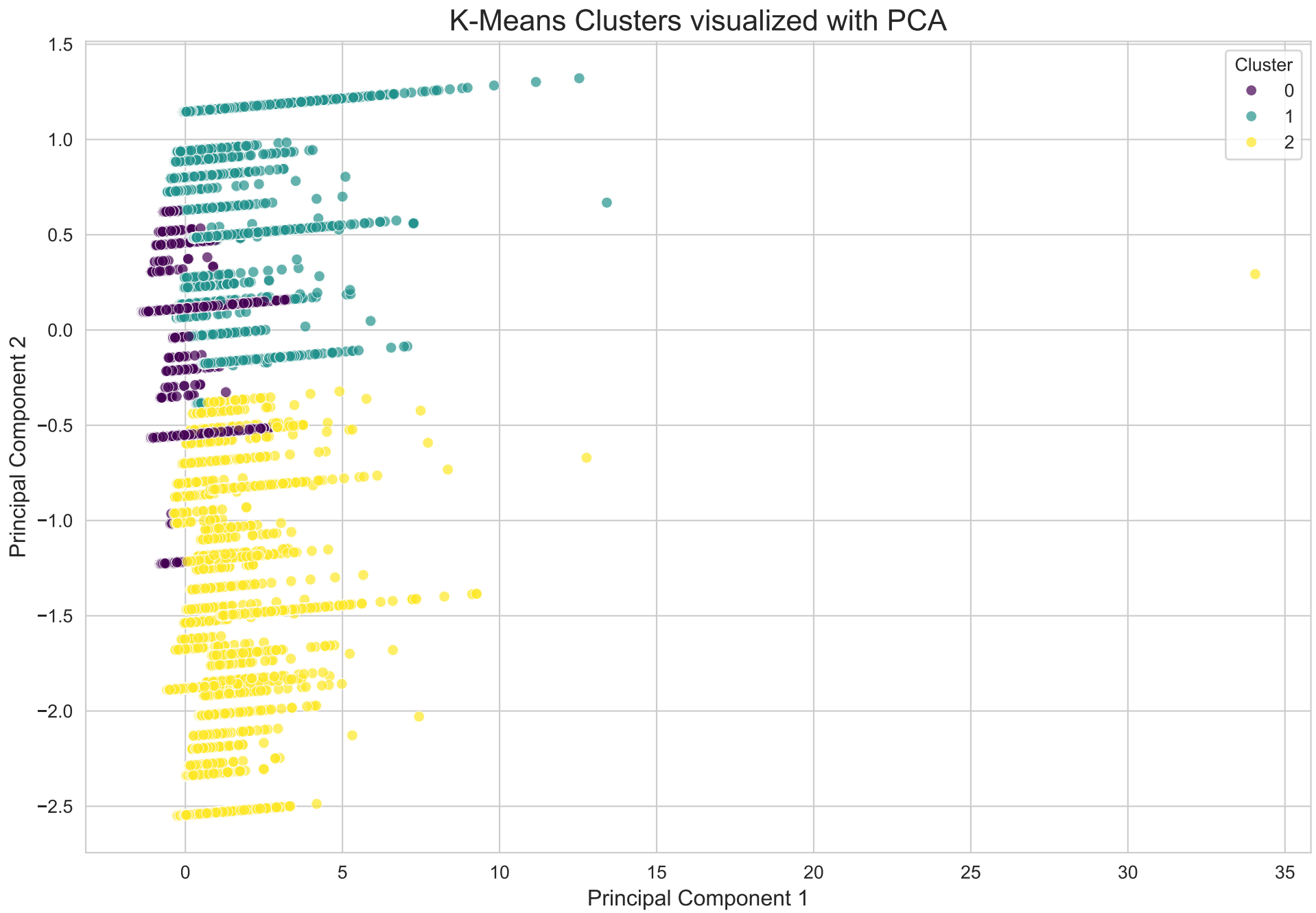

Helpfulness vs. Review Length
Score
1
2
3
4
5
Helpfulness Numerator
0
1
2
3
4
5
Review Length
0
1000
2000
3000
4000
5000


Top 10 Most Reviewed Products
B007JFMH8M
B002QWHJOU
B0026RQTGE
B002QWP8H0
B002QWP89S
B003B3OOPA
B001EO5Q64
B0013NUGDE
B007M83302
B001RVFEP2
Product ID
0
200
400
600
800
Number of Reviews


Average Score Trend Over Years
Average Score
5.0
4.8
4.6
4.4
4.2
4.0
3.8
3.6
2000
2002
2004
2006
2008
2010
2012
Year

Most Common Words in Reviews


Random Forest Confusion Matrix
Actual
negative
neutral
positive
1459
0
0
0
745
0
0
0
7796
negative
neutral
positive
Predicted
7000
6000
5000
4000
3000
2000
1000
0


K-Means Clusters visualized with PCA
Cluster
0
1
2
Principal Component 2
Principal Component 1
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
0
5
10
15
20
25
30
35